Finance Flexee V3

CF Manufacturing Simulation
Complete Admin, Faculty & Student Workflow

Functional Workflow & UAT Draft

This document provides the complete end-to-end workflow of the CF Manufacturing Simulation. The system is documented through three separate user flows: **Admin Flow, Faculty Flow, and Student Flow**. It also explains how these flows connect through quarters, decisions, accounting elections, engine calculation, validation and financial reporting.

1. Overall Simulation Architecture

The simulation operates as a controlled quarter-by-quarter process. Admin establishes the course and configuration; Faculty operates the simulation and controls quarter progression; Students make firm-level decisions and review their financial results.

```
ADMIN → COURSE / SIMULATION SETUP → FACULTY → QUARTER MANAGEMENT → STUDENT → DECISIONS / ACCOUNTING ELECTIONS → FACULTY REVIEW & CLOSE → ENGINE → VALIDATION → RESULTS → NEXT QUARTER
```

2. Flow 1 — Admin

Step	Activity	Function
1	Login	Admin signs in to the system.
2	Course Management	Create or manage the simulation course.
3	Scenario / Simulation Setup	Configure the simulation scenario and required course settings.
4	Users / Access	Configure or manage faculty and student participation as applicable.
5	Accounting Election Control	Enable or disable Student Accounting Election Control at course level.
6	Dates / Configuration	Confirm simulation dates, quarter structure and required configuration.
7	Deploy	Deploy the configured course for faculty/student use.

Admin Flow Summary

```
LOGIN → CREATE / MANAGE COURSE → CONFIGURE SCENARIO → CONFIGURE USERS → CONFIGURE ACCOUNTING ELECTION CONTROL → CONFIRM DATES / SETTINGS → DEPLOY COURSE
```

3. Flow 2 — Faculty

Faculty operates the simulation after deployment, manages quarters, reviews accounting-election requests, starts the close process and confirms successful firm completion.

```
FACULTY LOGIN → SELECT COURSE → OPEN QUARTER → MONITOR STUDENT ACTIVITY → REVIEW ACCOUNTING ELECTIONS → START CLOSE → MONITOR FIRM CLOSES → QUARTER CLOSED → NEXT QUARTER
```

3.1 Faculty Course / Quarter Setup

• Sign in as Faculty.

• Open the assigned simulation course.

• Review course and simulation configuration.

• Confirm the Accounting Election Control setting.

• Open the next available quarter when ready.

3.2 Quarter Lifecycle

```
LOCKED → OPEN → PROCESSING → CLOSED
```

Locked: Students cannot enter/edit the quarter.

Open: Students can enter, save and submit decisions.

Processing: Faculty has started close; queued jobs calculate each participating firm.

Closed: All firms completed successfully and results are read-only.

3.3 Faculty Accounting Election Review

• View requests for the specific firm and quarter.

• Review submitter/member information and proposed methods.

• Approve or reject a request.

• When multiple pending requests exist for the same firm and quarter, first-approved-wins automatically rejects the other pending requests.

• Approved settings become effective for that firm and quarter.

3.4 Faculty Close Process

```
START CLOSE → COURSE PROCESSING → FIRM CLOSE JOBS → DECISION RESOLUTION → ENGINE CALCULATION →
FINANCIAL VALIDATION → PERSIST RESULT → FIRM SUCCESS → ALL FIRMS SUCCESSFUL → COURSE CLOSED
```

If a firm fails calculation or validation, the quarter does not incorrectly close. The failed firm must complete a successful close before the quarter can become CLOSED.

4. Flow 3 — Student

The Student works at the firm level, entering operational decisions for the current quarter and, when enabled, proposing accounting-policy changes for faculty approval.

```
STUDENT LOGIN → MY COURSES → OPEN SIMULATION → CURRENT QUARTER → ACCOUNTING ELECTIONS (IF ENABLED)
→ OPERATIONAL DECISIONS → SAVE / REVIEW → SUBMIT → FACULTY CLOSE → LOCKED → RESULTS → NEXT QUARTER
```

4.1 Student Login and Course Entry

• Student signs in.

• Student opens My Courses.

• Student selects the assigned simulation course.

• Student opens the current quarter.

4.2 Student Accounting Elections

When Student Accounting Election Control is ON, the student has a dedicated Accounting Elections page containing Current Accounting Settings, Accounting Settings and Accounting Election Request History.

Selectable methods:

• Depreciation Method

• Shipping Terms

• Bad Debt Method

• Warranty Method

• Excise Tax Treatment

Inventory Method: FIFO is fixed and cannot be changed.

4.3 Student Default Accounting Path

```
CURRENT EFFECTIVE SETTINGS → USE DEFAULT ACCOUNTING SETTINGS → NO FACULTY APPROVAL REQUIRED →
CONTINUE
```

4.4 Student Accounting Change Path

```
CURRENT SETTINGS → CHANGE ACCOUNTING SETTINGS → SELECT METHODS → SUBMIT REQUEST → PENDING → FACULTY REVIEW → APPROVED / REJECTED
```

Pending: The proposal is stored but does not affect Current Accounting Settings or the engine.

Approved: The proposed settings become effective for the specific firm and quarter.

Rejected: The previous effective settings remain unchanged and the request remains in history.

4.5 Multiple Student Members

Multiple members of the same firm may submit accounting-election requests for the same quarter. Only one can become effective; approval of one automatically rejects other pending requests for that firm and quarter.

4.6 Student Operational Decisions

• Forecasting — P1/P2 demand by region and confidence.

• Sourcing — primary supplier, allocation and secondary supplier.

• Production — P1/P2 production, shifts and postponed-production settings.

• Marketing — budget percentage and segment allocations.

• Quality — inspection level.

• Logistics — logistics mode.

• Certification — required statement attestation.

• Cash Dividend — dividend amount within the available limit.

4.7 Save Draft and Submit

```
SAVE DRAFT → REVIEW → SUBMIT FINAL → SUBMITTED / EDITABLE UNTIL FACULTY STARTS CLOSE
```

Once Faculty begins closing the quarter, the decision becomes locked and cannot be changed.

5. What Happens After Student Submission

Student submission does not immediately create final financial results. The submitted decision waits for the faculty-controlled quarter close.

```
STUDENT SUBMISSION → FACULTY CLOSE → LOCK DECISION → RESOLVE COMPLETE DECISION SET → CALCULATE
```

6. Decision Resolution

At close time, the system resolves the complete decision payload from operational decisions, system-resolved values, certification and the effective accounting election for the firm and quarter.

When Accounting Election Control is ON, accounting settings are supplied as effective accounting settings and are not treated as ordinary operational Decisions.

7. Manufacturing Engine

The Manufacturing Engine calculates the quarter using the opening financial position, resolved decisions, simulation parameters, previous-quarter information and effective accounting settings.

• Income Statement — revenue, COGS, gross profit, operating expenses, operating income, interest, tax and net income.

• Balance Sheet — cash, receivables, inventory, PPE, payables, debt, retained earnings and other balances.

• Cash Flow Statement — operating, investing and financing cash flows and ending cash.

• Supporting engine fields and intermediates used for reporting and validation.

8. Financial Validation

The calculated result is validated before persistence. Checks include balance-sheet balance, retained-earnings coherence, working-capital relationships, turnover metrics, cash movement and quarter-over-quarter plausibility.

Zero-activity quarters are supported. When revenue, supplier purchases and production are zero, activity-dependent plausibility checks are treated as not applicable where appropriate.

9. Quarter Result Persistence

After successful calculation and validation, the firm's engine result, resolved decision payload and close information are persisted. The firm close attempt becomes successful.

The course quarter is CLOSED only after all participating firms have successfully completed their close attempts.

10. Student Financial Dashboard

After closure, students can view finalized financial history in read-only mode.

• Quarter-over-quarter trend.

• Net Revenue.

• Cash.

• Equity.

• Stored Operating Margin.

• Income Statement.

• Balance Sheet.

• Cash Flow Statement.

• Persisted reports and notes.

11. Quarter-to-Quarter Accounting Inheritance

If a later quarter has no own approved election, the latest previous effective election is inherited. A later quarter may still receive a new proposal; once approved, that new election becomes effective for that quarter.

```
Q4 APPROVED → Q5 NO NEW APPROVED ELECTION → Q5 INHERITS Q4 → Q6 NO NEW APPROVED ELECTION → Q6 INHERITS Q5
```

12. Accounting Election ON vs OFF

Area	Control ON	Control OFF
Student Accounting Elections	Available	Disabled / greyed out
Accounting choices in normal Decisions	Removed	Existing/default behavior
Student proposal	Available	Not available
Faculty approval	Required for changes	Not applicable
Engine accounting settings	Effective firm/quarter election	Existing/default accounting behavior

13. Complete Role-Based Workflow

ADMIN

```
LOGIN → COURSE → SCENARIO → USERS → ACCOUNTING CONTROL → CONFIGURATION → DEPLOY
```

FACULTY

```
LOGIN → COURSE → OPEN QUARTER → MONITOR → ACCOUNTING REVIEW → APPROVE/REJECT → START CLOSE → MONITOR → CLOSE
```

STUDENT

```
LOGIN → COURSE → QUARTER → ACCOUNTING ELECTIONS → OPERATIONAL DECISIONS → SAVE/SUBMIT → LOCK → RESULTS → NEXT QUARTER
```

14. Responsibilities Matrix

Responsibility	Admin	Faculty	Student
Course creation / management	Yes	As permitted	—
Scenario / simulation configuration	Yes / authorized	Yes / authorized	—
User / participation setup	Yes / authorized	As permitted	—
Enable Accounting Election Control	Yes / course administration	Yes / authorized	—
Deploy course	Yes / authorized	Yes / authorized	—
Open / manage quarters	Depending on role	Yes	—
Enter operational decisions	—	—	Yes
Submit accounting proposal	—	—	Yes
Approve / reject accounting proposal	—	Yes	—
Start quarter close	—	Yes	—
Review finalized results	Administrative access	Yes	Own firm
View finalized dashboard	—	Yes	Yes

15. Final UAT Checklist

• Admin login and course-management permissions work.

• Faculty login and quarter-management permissions work.

• Student can access the assigned course and current quarter.

• Accounting Election Control ON/OFF behavior is correct.

• Student can use default accounting settings.

• Student can submit an accounting-change proposal.

• Pending proposals do not change Current Accounting Settings.

• Faculty approval and rejection work correctly.

• First-approved-wins works for multiple pending requests.

• Later-quarter inheritance works.

• Later-quarter approved requests override inherited settings.

• Operational decisions save and submit correctly.

• Submitted decisions lock when faculty starts close.

• All firms must close successfully before the quarter becomes CLOSED.

• Engine uses the resolved accounting settings.

• Financial validation runs before persistence.

• Zero-activity quarters are handled correctly.

• Closed-quarter dashboard results are read-only and show persisted values.

16. Current Implementation Status

• Course-level Accounting Election Control implemented.

• Student Accounting Elections UI implemented.

• Default/change accounting paths implemented.

• Five selectable accounting methods implemented; FIFO fixed.

• Pending, approval and rejection workflow implemented.

• First-approved-wins implemented.

• Firm + quarter-specific effective accounting resolution implemented.

• Quarter inheritance implemented.

• Accounting elections integrated into decision resolution and Manufacturing Engine.

• Accounting control OFF regression preserved.

• Zero-activity plausibility handling implemented and tested.

• Quarter close, persistence and read-only dashboard flow tested.

Document status: Draft for functional documentation and UAT. Update after final Admin, Faculty and Student acceptance testing.